

Luigui Gallardo-Becerra

BIOINFORMATICIAN | MOLECULAR BIOLOGIST

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Experienced Bioinformatician and Molecular Biologist seeking new challenges to apply and expand my expertise in computational biology, data science, and molecular research. Over the past several years, I have contributed to diverse research projects utilizing modern genomic technologies (NGS), advanced data analysis, and data science methodologies to generate novel biological insights that have resulted in high-impact peer-reviewed scientific publications.

Research Experience

Graduate Research Assistant (Bioinformatics & Molecular Biology)

Institute of Biotechnology, UNAM

January 2019 - July 2025

Across several interdisciplinary research projects, I contributed to data acquisition, computational analysis, and dissemination of scientific results. My responsibilities included:

Project conceptualization: Assisted in defining research goals, experimental design, analytical workflows, and computational strategies to address multi-omics biological questions.

HPC system administration: Maintained laboratory computational infrastructure, including Linux installation, software configuration, performance optimization, and data security practices.

Reproducible workflow engineering: Built robust, scalable, and reproducible pipelines using Snakemake, Python, Bash, and Rabix Composer to automate NGS and metagenomic analyses.

Custom tool development: Implemented specialized algorithms and utilities in Python, R, Bash, and Perl to support complex multi-step analyses.

Quantitative data visualization: Designed comprehensive visualizations and figure panels using ggplot2, seaborn, matplotlib, GraphPad, Tableau, and Jupyter Notebooks.

Computational best practices: Maintained version control through GitHub repositories and incorporated CI/CD testing to ensure workflow reliability.

Scientific reporting: Prepared internal research reports, contributed to manuscript writing, and provided analytical support for high-impact publications.

Presentation & outreach: Delivered oral and poster presentations at national and international conferences and participated in science communication initiatives.

Education

National Autonomous University of Mexico (UNAM)

PH.D. IN COMPUTATIONAL BIOLOGY

Mexico City, Mexico

January 2019 - July 2025

National Autonomous University of Mexico (UNAM)

MASTER OF SCIENCE IN COMPUTATIONAL BIOLOGY

Mexico City, Mexico

August 2016 - January 2019

University of Guadalajara (UDG)

BACHELOR OF SCIENCE IN MOLECULAR BIOLOGY

Guadalajara, Mexico

August 2012 - January 2016

Teaching Experience

Guest lecturer at Universidad de La Sabana (Chia, Colombia). December 2023.

Microbiota Symposium from Theory to Clinical Practice, Colegio de Profesionales de la Nutrición de Querétaro y el Bajío. June 2021.

Bioinformatics and Reproducible Research Workshop, Postgraduate Program in Biological Sciences, UNAM. October 2019.

Guest lecturer at Center of Genomic Sciences, UNAM. August 2018.

Directed Theses

Bachelor's Thesis: Itzel Abigail Hernández Reyna, Bachelor's in Nutrition (Universidad Autónoma del Estado de Morelos), December 2019. "Impact of bacteriophages associated with childhood obesity on the intestinal metagenome."

Soft Skills

Adaptable and quick to learn new technologies, analytical methods, and computational tools.

Strong problem-solving skills with the ability to manage multiple complex projects while maintaining high-quality output.

Effective communicator with experience collaborating in interdisciplinary teams as well as working independently.

Skilled in technical operations including server maintenance, Git/GitHub workflows, and development of automated scripts for research tasks.

Programming Languages, Tools, Bioinformatics & Molecular Biology Skills

Programming Languages:

Python, R, Bash, SQL, C#, Java, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries:

React, Django, ASP.NET Core, Shiny

Computing & DevOps:

Linux, High-Performance Computing (HPC/SLURM), Docker, Git/GitHub, CI/CD workflows, Conda

Data Science & Visualization:

ggplot2, seaborn, matplotlib, GraphPad, Tableau, Jupyter Notebooks, RStudio

Databases:

MySQL, PostgreSQL, MongoDB

Bioinformatics Tools & Methods:

Snakemake, Nextflow, QIIME2, Trinity, SPAdes, Bowtie2, BWA, HISAT2, samtools, fastp, Kraken2

NGS & Multi-omics Analysis:

RNA-seq, WGS, 16S rRNA profiling, metagenomics, metatranscriptomics, viromics

Specialized Bioinformatics Skills:

Genome and transcriptome assembly & annotation; differential expression analysis; taxonomic profiling; functional & pathway analysis; ETL automation for multi-omics data

Molecular Biology Skills:

DNA/RNA extraction, PCR/qPCR, gel electrophoresis, sample QC, NGS library preparation, bacterial culture & handling, plasmid preparation, sterile technique (BSL-2), and collaboration with wet-lab teams

Languages

English – Full professional proficiency

Spanish – Native

Publications

1. **Gallardo-Becerra L**, Cornejo-Granados F, Bikel S, Arenas I, López-Leal G, Alvarado-Gonzalez C, et al. Bioactive plasmid- and phage-encoded antimicrobial peptides (AMPs) in the human gut: a metatranscriptome–virome profiling reveals exploratory links to metabolic human diseases. *Microb Ecol*. 2025 Nov 28.
2. **Gallardo-Becerra L**, Cornejo-Granados F, Romero-Hidalgo S, Lopez-Zavala AA, Cota-Huizar A, Cervantes-Echeverría M, et al. Host genome drives the microbiota enrichment of beneficial microbes in shrimp: exploring the hologenome perspective. *Animal Microbiome*. 2025 May 22;7(1):50.
3. Manzo R, **Gallardo-Becerra L**, Díaz de León-Guerrero S, Villaseñor T, Cornejo-Granados F, Salazar-León J, et al. Environmental Enrichment Prevents Gut Dysbiosis Progression and Enhances Glucose Metabolism in High-Fat Diet-Induced Obese Mice. *International Journal of Molecular Sciences*. 2024 Jan;25(13):6904.
4. Jatuyosporn T, Laohawutthichai P, Romo JPO, **Gallardo-Becerra L**, Lopez FS, Tassanakajon A, et al. White spot syndrome virus impact on the expression of immune genes and gut microbiome of black tiger shrimp *Penaeus monodon*. *Sci Rep*. 2023 Jan 18;13(1):996.
5. **Gallardo-Becerra L**, Cervantes-Echeverría M, Cornejo-Granados F, Vazquez-Morado LE, Ochoa-Leyva A. Perspectives in Searching Antimicrobial Peptides (AMPs) Produced by the Microbiota. *Microb Ecol*. 2023 Dec 1;87(1):8.
6. Chino de la Cruz CM, Cornejo-Granados F, **Gallardo-Becerra L**, Rodríguez-Alegría ME, Ochoa-Leyva A, López Munguía A. Complete genome sequence and characterization of a novel *Enterococcus faecium* with probiotic potential isolated from the gut of *Litopenaeus vannamei*. *Microbial Genomics*. 2023;9(3):000938.
7. Cervantes-Echeverría M, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. The Two-Faced Role of crAssphage Subfamilies in Obesity and Metabolic Syndrome: Between Good and Evil. *Genes*. 2023 Jan;14(1):139.
8. Palomino-Hermosillo YA, Berumen-Varela G, Ochoa-Jiménez VA, Balois-Morales R, Jiménez-Zurita JO, Bautista-Rosales PU, et al. Transcriptome Analysis of Soursop (*Annona muricata* L.) Fruit under Postharvest Storage Identifies Genes Families Involved in Ripening. *Plants (Basel)*. 2022 Jul 7;11(14):1798.
9. Ochoa-Romo JP, Cornejo-Granados F, Lopez-Zavala AA, Viana MT, Sánchez F, **Gallardo-Becerra L**, et al. Agavin induces beneficial microbes in the shrimp microbiota under farming conditions. *Sci Rep*. 2022 Apr 16;12(1):6392.
10. Bikel S, **Gallardo-Becerra L**, Cornejo-Granados F, Ochoa-Leyva A. Protocol for the isolation, sequencing, and analysis of the gut phageome from human fecal samples. *STAR Protocols*. 2022 Mar 18;3(1):101170.
11. Bikel S, López-Leal G, Cornejo-Granados F, **Gallardo-Becerra L**, García-López R, Sánchez F, et al. Gut dsDNA virome shows diversity and richness alterations associated with childhood obesity and metabolic syndrome. *iScience*. 2021 Aug 20;24(8):102900.
12. **Gallardo-Becerra L**, Cornejo-Granados F, García-López R, Valdez-Lara A, Bikel S, Canizales-Quinteros S, et al. Metatranscriptomic analysis to define the Secrebiome, and 16S rRNA profiling of the gut microbiome in obesity and metabolic syndrome of Mexican children. *Microb Cell Fact*. 2020 Dec;19(1):61.
13. **Gallardo-Becerra L**, Cornejo-Granados F, Leonardo-Reza M, Ochoa-Romo JP, Ochoa-Leyva A. A meta-analysis reveals the environmental and host factors shaping the structure and function of the shrimp microbiota. *PeerJ*. 2018 Aug 10;6:e5382.
14. Cornejo-Granados F, Lopez-Zavala AA, **Gallardo-Becerra L**, Mendoza-Vargas A, Sánchez F, Vichido R, et al. Microbiome of Pacific Whiteleg shrimp reveals differential bacterial community composition between Wild, Aquacultured and AHPND/EMS outbreak conditions. *Sci Rep*. 2017 Dec;7(1):11783.